CLIAHUB How-To Document

Exec Summary:

This document describes how the CZ Biohub rapidly converted from a Research Only mission to a CLIA validated clinical testing operation, in the span of one week. This rapid conversion and deployment in the face of a global pandemic emergency necessarily required significant multidisciplinary inputs, including a new organizational structure, volunteers, automation, just in time engineering, manufacturing, software engineering, molecular biology, and regulatory approval. All aspects of the successful conversion to clinical testing will be discussed, including strengths and vulnerabilities, with the goal of providing a lasting roadmap for other research organizations, and future deployments.

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Introduction

Accurate diagnostic testing is a pillar of outbreak response, allowing clinicians and public health practitioners to effectively identify infected individuals to inform treatment and public health interventions, such as quarantine. Emerging into the human population sometime in late November 2019, the SARS-CoV-2 pandemic has further highlighted the foundational importance of diagnostic testing with multiple studies indicating that asymptomatic and presymptomatic individuals are drivers of community transmission [REF]. Thus, a broad testing capability will be a necessary component in any strategy that seeks to both control new outbreaks of the disease and re-open the global economy.

Despite local and state public health laboratories across the United States having operationalized a diagnostic molecular assay for SARS-CoV-2 infection in early February, limited testing capacity and uneven availability of testing significantly hampered the early response to the pandemic in the United States. These limitations resulted in tighter criteria for testing, including prioritization for hospitalized patients, healthcare workers, nursing homes, and other high-risk populations. Poorly resourced public health laboratories, lack of qualified personnel, scarcity of reagents, swabs, PPE, and other supplies compounded the problem, ultimately limiting the ability to stop or slow new introductions into and throughout the United States.

Laboratory barriers to testing are the limited availability of space, high capital infrastructure (robotics, information systems), material resources with uninterrupted supply chains, and trained

personnel qualified to conduct assays in Clinical Laboratory Improvement Amendment (CLIA)-approved laboratories. However, sizable untapped capacity exists throughout the United States in a multitude of university research laboratories. Theoretically, this capacity could be directed to rapidly expand emergency response capabilities; however, real barriers exist to their entry into clinical diagnostic testing. Research laboratories -- while working with the same suite of methods and from the same biological principles of clinical diagnostic laboratories -- have substantively different workflows, expectations of quality control and validation, information management, documentation, traceability, reporting, and regulatory compliance. For research scientists, this burden may appear extremely formidable.

To address the dearth testing availability in the San Francisco Bay Area, the Chan Zuckerberg Biohub, in partnership with the UCSF Clinical Microbiology Lab, rapidly built and deployed an emergency COVID-19 viral testing facility, staffed by a combination of Biohub employees and UCSF graduate students, postdocs, and faculty. These efforts were facilitated by an Executive Order (March 12, 2020) from California State Governor Gavin Newsom modifying certain California-specific regulatory personnel requirements. The goal of this facility, called the "CLIAHUB," was to provide free, rapid, high capacity testing services to local hospitals, Departments of Public Health offices, and community-based clinical screening in the San Francisco Bay Area, with a particular emphasis on underserved populations. In this perspective piece, and in the accompanying document (Link-out: "The CLIAHUB: How-to Guide for Building a CLIA Lab for SARS-CoV-2 Testing"), we describe the experiences of the Biohub, including our organizational structure, recruitment and training of highly-skilled volunteers, automation, information management, engineering, protocol development, and clinical validation. We share our lessons learned and offer guidance to other research laboratories to provide a model for rapid expansion of diagnostic capabilities in the context of emergency response for COVID-19 and future pandemic responses.

The Chan Zuckerberg Biohub is a non-profit research institute that employs approximately 110 people organized into multiple smaller research groups, with nearly half of the staff focused on infectious disease. The organization's strong focus on collaborative work, communication and open science, allowed frictionless redistribution of personnel priorities into COVID-19 work. On March 12th, the Biohub's COVID-19 response staff was organized into thirteen work teams (Figure 1A). Staff members from various research groups and seniority levels were appointed as Team Leads and Backup Team Leads according to their skills and experience. To ensure continuity in the event of an outbreak, all job functions and critical knowledge were held by at least two individuals. Communications were facilitated by daily All Hands meetings and dedicated Slack channels. Once clinical testing was underway, volunteers from UCSF were trained to run the assay and organized into four non-overlapping squads to reduce viral transmission risk.

Clinical laboratories are required to meet all applicable federal, state and local regulations in order to perform patient testing and have to be certified with their state and the Center for

Medicare and Medicaid Services (CMS) under CLIA. These regulations specify the laboratory organizational structure and minimum operational standards, with major areas covering the laboratory environment, staffing roles and requirements, information tracking and management, testing validation and procedures, quality assurance, and proficiency testing. Each of these areas had to be addressed with compliance documentation as the CLIAHUB was formed. Ensuring regulatory compliance required close collaboration between the CLIA laboratory director, microbiology section director, clinical quality assurance and Biohub laboratory lead staff. Since the CLIA laboratory director is ultimately responsible for the operations and quality of test results reported by the laboratory, the importance of this close partnership between the UCSF clinical laboratory and the Biohub cannot be overstated.

The overall workflow is shown in Figure 1B. Agilent Bravo liquid handlers and Biorad CFX384 qPCR machines were selected due to local availability, open supply chain, strong technical support, and familiarity among the staff. The Viral RNA Magbead Kit from Zymo Research and the Luna qPCR Master Mix were chosen based on reagent availability after significant investigation by the Supply Chain team. An initial experiment comparing various established Taqman assays led to selection of one N and one E gene probe with the best performance. Based on stability studies and safety concerns, DNA/RNA Shield was chosen for the sample collection media. To supply local DPHs with thousands of collection tubes containing Shield, a custom tube-filling robot was engineered. The transfer of sample from collection tube to 96 well plate posed a challenge, which was solved in two ways in parallel: manual transfer with a custom built LED guided system, and fully automated, using a Hamilton STARlet liquid handling system. Another critically important component was a Laboratory Information System (LIS) capable of seamlessly tracking reagents, the acquisition of samples, their placement into plates, and their associated results. Biohub data scientists along with software engineers from the Chan Zuckerberg Initiative chose to build an agile and serverless LIS which is described in detail in the accompanying document. After assay optimization, limit of detection (LOD) was established by serial dilution of a quantified high titer patient sample, and validation of the assay for the EUA application was achieved by re-testing samples previously identified as positive or negative based on a previously approved assay at the UCSF clinical lab.

On March 20th, 8 days after starting this project, the CLIAHUB returned its first clinical result. As of this writing, the CLIAHUB has returned XX,XXX clinical results. A total of XX California DPHs, safety net hospitals, jails, skilled nursing facilities, and public clinics have utilized the CLIAHUB for SARS-CoV-2 testing, in addition to community-based studies. In the period from May 8th to June 8th, the CLIAHUB processed XXXX on average per day. Turn around time was XX hrs for the majority of samples. For genomic epidemiology, positive samples were then routed for full SARS-CoV-2 genome sequencing. While the samples processed by CLIAHUB were heterogeneous and not representative of the general population, several important characteristics of the disease and the test characteristics were notable. A broad range of Ct values have been observed (from 7 to 45, Figure 2) is indicative of a tremendous range of viral loads within this disease. While there was excellent agreement between both viral probes, in

X% of cases, only one of the two probes were detected, suggesting that assays that rely on detection of a single gene alone may have a significant false negative rate. Furthermore, we observe a large number of high Ct samples suggesting that pooling strategies would likely result in a decrease in assay sensitivity, and an increase in false negatives.
[1250 words]

The largest users of CLIAHUB testing capacity have been Bay Area county DPHs, including San Mateo, Marin, Alameda, Placer, Tehama, Santa Clara, and Sonoma counties. Feedback from individual counties indicated that access to free and fast testing allowed insight into at-risk populations, which in turn led to deployment of medical personnel. The case of congregate care facilities, intervention and stabilization avoided evacuation of residents to local hospitals. The CLIAHUB also directly enabled community based studies, such as studies within the town of Bolinas, the Mission district of San Francisco, Bayview Hunters Point, and a study of unsheltered persons. These studies provided both clinically actionable results for individuals and their families, in addition to valuable insight about the socioeconomics and demographics of populations that were disproportionately affected. All together, the experience of CLIAHUB demonstrates that research facilities can indeed be converted to clinical testing laboratories that are capable of returning high quality and timely clinical results in order to meet the challenge of a growing pandemic. We hope that the accompanying “How to” document, protocols, and engineering details will serve as a guide for others in the midst of this, or any other future pandemic.

Figure 1a

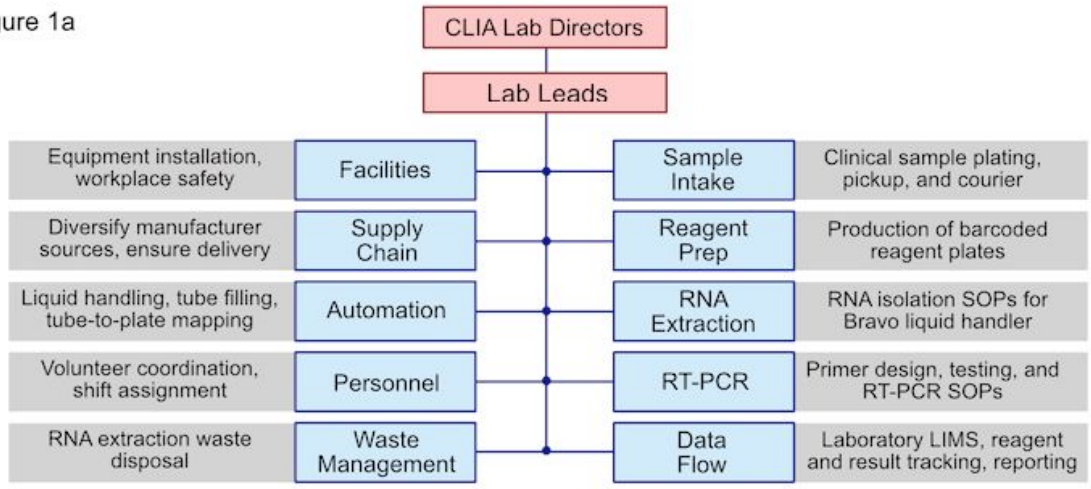
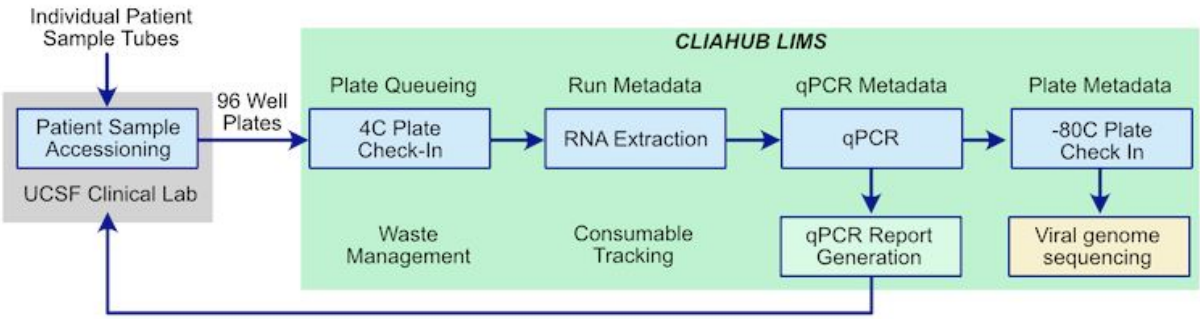


Figure 1b



Work Flow and Design Specifications

Teams

A. Regulatory Requirements

Clinical laboratories are required to meet all applicable federal, state and local regulations in order to perform patient testing and have to be certified with their state and the Center for Medicare and Medicaid Services (CMS) under the Clinical Laboratory Improvement Amendments (CLIA). These regulations specify the laboratory organizational structure and minimum operational standards, with major areas covering the laboratory environment, staffing roles and requirements, information tracking and management, testing validation and procedures, quality assurance, and proficiency testing. Each of these areas had to be

addressed with compliance documentation. The California Executive Order on March 11, 2020 allowed for certain state requirements to revert to the federal level, notably enabling 1) designation of an off-site location as part of the UCSF clinical laboratories for SARS-CoV-2 testing and 2) use of testing personnel meeting federal training and competency requirements but not needing a state-issued Clinical Laboratory Scientist (CLS) license. Applying the clinical laboratory organizational structure and ensuring regulatory compliance required close collaboration between the CLIA laboratory director, microbiology section director, clinical quality assurance and CZ Biohub laboratory lead staff. Since the CLIA laboratory director is ultimately responsible for the operations and quality of test results reported by the laboratory, the importance of this close partnership between the UCSF clinical laboratory and CZ Biohub cannot be overstated.

During the COVID-19 outbreak, the FDA initially developed a policy requiring EUA approval prior to performance of SARS-CoV-2 testing. This was changed on February 29, 2020 to allow for testing after laboratory validation with submission of EUA within 15 days of beginning testing. The laboratory validation was performed and submitted for authorization based on FDA guidance with performance characteristics determined for analytical limits of detection, clinical sample evaluation, and in silico analysis for inclusivity and cross-reactivity. Subsequent communications with the FDA provided additional validation data and assay revisions to deal with supply shortages and improve performance. Quality assurance activities included review of plate data prior to results reporting, sample result correlation, repeating of samples with unusual results, and monitoring of control performance and sample issues.

B. Supply Chain

Supply chains worldwide have been affected due to the COVID-19 pandemic. To ensure the lab had all the supplies needed for COVID-19 testing and research, we organized a team of 6 to 9 people, distributed into different team functions: Ordering, Order Tracking, Receiving and Inventory. Each team member had defined roles with built-in redundancies. Team meetings were held twice-daily to address any supply chain disruptions.

A primary list of items needed to process an estimated 100,000 clinical samples was specified in conjunction with the leads of the other teams. Supplies specified by published COVID SOPs were subject to shortages, and therefore, alternatives were strategically sourced. This included preferred vendors offering discounts and substitute items for alternative workflows and contingencies. The supply chain team worked closely with vendors to determine availability of items prior to ordering, and tracked shipments, sometimes hourly, to ensure delivery. Common problems and disruptions included: stuck and lost items, backordered items, and glitches in the courier delivery systems. Effective long-standing business partnerships with vendors and sales representatives were critical in resolving these problems during this time of need and uncertainty, while also securing discounted pricing for large orders.

In some cases, the assay development teams pursued the use of reagents from multiple sources, while the supply team worked to secure sufficient quantities of each. For example, within the first week the RNA extraction team developed parallel protocols using Thermo Fisher's MagMAX kit and Zymo Research's Viral RNA MagBead kit. Similarly, the PCR team piloted Thermo Fisher's TaqPath qPCR kit and New England Biolab (NEB)'s Luna qPCR kit. Within 1-2 weeks it had become clear that Thermo Fisher's supply chains were struggling while Zymo and NEB were able to supply sufficient quantities of reagents for 100,000 tests. This may be due to the fact that Thermo Fisher's reagents had been used in the CDC's first SARS CoV2 assay and thus were in extremely high demand from clinical labs around the world. Zymo and NEB primarily supply the research industry, and so were less sought-after in the early days of March 2020. Under CLIA, the UCSF Clinical Lab, of which CLIAHUB is a part, is able to use non-FDA-approved reagent in the context of a site-specific Laboratory Developed Test (LDT) as long as it meets certain performance characteristics, and in this case use of non-traditional reagents with more secure supply chains was critical.

For all received orders, the contents were verified and logged into the inventory system. Once supplies began to arrive and throughout the rest of the CLIAHUB's time, the Supply Chain team conducted bi-weekly inventory of items to ensure that there were sufficient quantities, and was also charged with informing the research teams when supplies were running low.

C Equipment and Facilities

To supplement the testing throughput of UCSF's Clinical Diagnostic Laboratory, an auxiliary diagnostic lab was rapidly built at Mission Bay by Chan Zuckerberg Biohub staff and UCSF staff. The lab and support spaces cover 6,827 square feet of research laboratories leased by UCSF , consisting of an open lab area plus six bays. Essential facility features include: 4C and -20C coldrooms, gowning rooms, shipping/receiving, waste storage, and facilities room. A suite of BSL-2 rooms (1,969 sqft) were repurposed for reagent preparation, collection tube filling, and reagent aliquoting (under positive pressure). A separate space located in a different part of the building was used to prepare positive control samples to reduce the risk of contamination.
Layout: Sample processing consists of 8 identical stations, each consisting of an Agilent Bravo liquid handler, a real-time qPCR machine (Bio-Rad CFX384), and PCs to run each. To optimize personnel flow, no two stations were placed back-to-back. All stations are situated either linear (covering 11 linear feet of bench space), or the Bravo and qPCR sections are across from one another (each covering 6 linear feet of bench space). [FIGURE]
Equipment: Additional equipment includes 4 refrigerated plate and tube centrifuges; 2 PCR hoods; 4C refrigerator for incoming sample storage; additional cold storage at 4C, -20C, -80C; walk-in cold storage at 4C, -20C; flammable storage for prepared reagent plates. A Hamilton STARlet was installed in a separate room for transfer from collection tubes to 96 well deep well plates; a second one was eventually installed as well. A ducted BSC is used for aliquoting noxious liquids such as viral prep buffer, which contains BME.
Safety: Lab workers enter through the gowning room, where all personnel don PPE. Glove boxes are located at multiple points throughout the lab.
Equipment Acquisition: Multiple Bravo and Bio-Rad qPCR machines were redeployed from neighboring UCSF laboratories and local biotech companies. In particular, Zymogen and Agilent

provided both equipment and automation engineers to train and assist with machine calibration and programming. Generous consumable donations from multiple nearby organizations enabled the lab to get up and running within days.

Power: The lab space is powered by standard outlets in wiremold around the perimeter, connected to 20-amp circuits at the panels, including multiple outlets connected to E-Power backup provided by a 750kW generator. In the event of a power outage, this generator engages and supplies power. UPS units were installed to supply power continuously and prevent loss of any samples on the Bravos, which would reset in the event of a momentary power loss.

Janitorial: In addition to the regular duties performed by the janitorial crew in the office and lab, the crew cleans all door handles, conference room tables and chairs, common area tables and chairs, kitchen surfaces and appliances with disinfectant wipes and cleaner nightly. Each restroom has a cell phone cleaning station. The day porter was also charged with disinfecting all of the door handles multiple times per day.

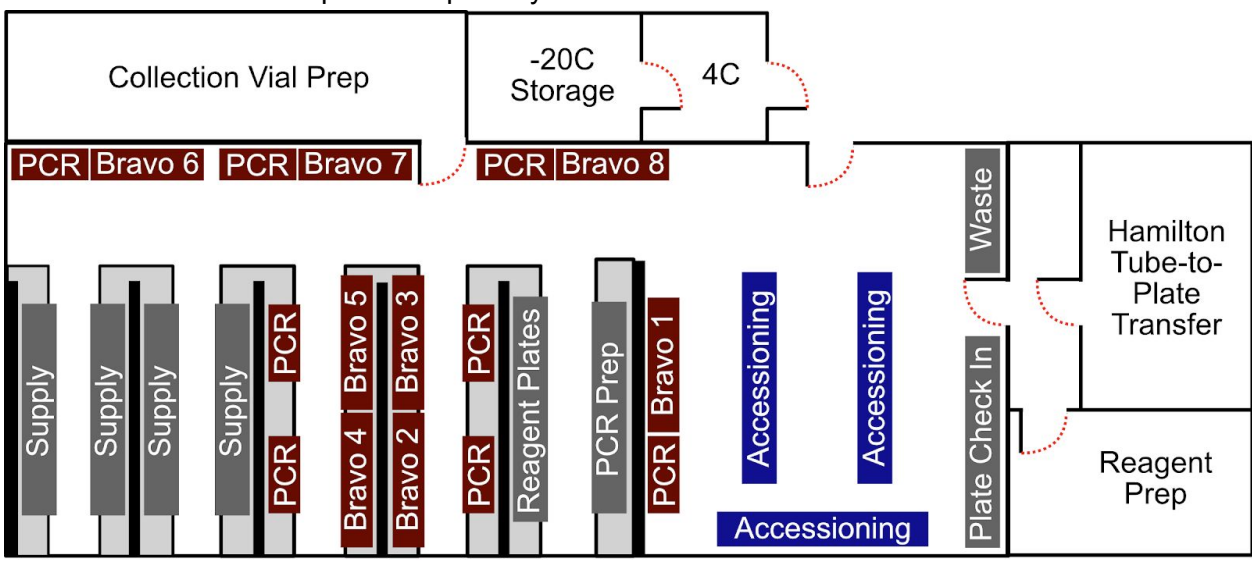


Figure XX. Facilities floor plan for CLIAhub. Clinical collection vials are first accessioned (Blue) and barcoded for subsequent transfer into 96 well plates as described in Section E, below. Plates are then distributed to each of the 8 extraction and PCR stations (Red). Additional rooms were used for reagent preparation and collection vial preparation as described below.

D. Tubes and swabs

A hindrance to more widespread testing was the availability of both nasopharyngeal swabs and sample collection tubes, typically filled with Universal Transport Media (UTM) or Viral Transport Media (VTM). In some cases, health care providers possessed swabs, but lacked the collection tube itself. We addressed these issues by directly sourcing NP swabs from a manufacturer (iClean, Shenzhen, China) and producing tubes that contain Zymo DNA/RNA Shield. Shield is compatible with most RNA extraction procedures and kits, preserves nucleic acids, and inactivates pathogens, thus providing protection for personnel (CDC compliant and EuroVir certified for pathogen inactivation) such that all downstream workflows can be processed in a

BSL1 lab environment. Validation of viral RNA stability in Shield relative to saline is described in section J. The iClean swabs have 80mm breakpoints so we secured a supply of tubes appropriate for these swabs. Tubes that are too short result in compressed swabs that can splash and generate aerosols during deswabbing. Tubes that are too tall make it difficult to remove swabs. Additionally, we sought tubes that would be compatible with our automation platforms (see below). We decided to use 16x79mm polypropylene tubes (Sarstedt 60.551) because of their compatible size and availability.

To simplify and accelerate the process of Shield tube filling, a mechanical dispenser, (nicknamed “the Bartender”, **Figure XX**) was created to automate the aliquoting process. The Bartender dispenses into batches of 96 tubes located on swappable racks to minimize downtime. A single Shield tube production line can produce 2,000 tubes per hour. The Bartender was designed to operate inside standard biosafety cabinets and be adjustable to adapt to collection tubes of different heights and diameters. Individual tubes were labeled with identification information as well as an expiration date. Because Shield is a guanidinium-containing reagent which is incompatible with bleach, a warning was added to the tube labels to reduce confusion in case Shield tubes were accidentally sent to a different processing lab. As a quality control measure, boxes of tubes were periodically sampled and processed in batches through the standard clinical testing workflow to assess contamination.

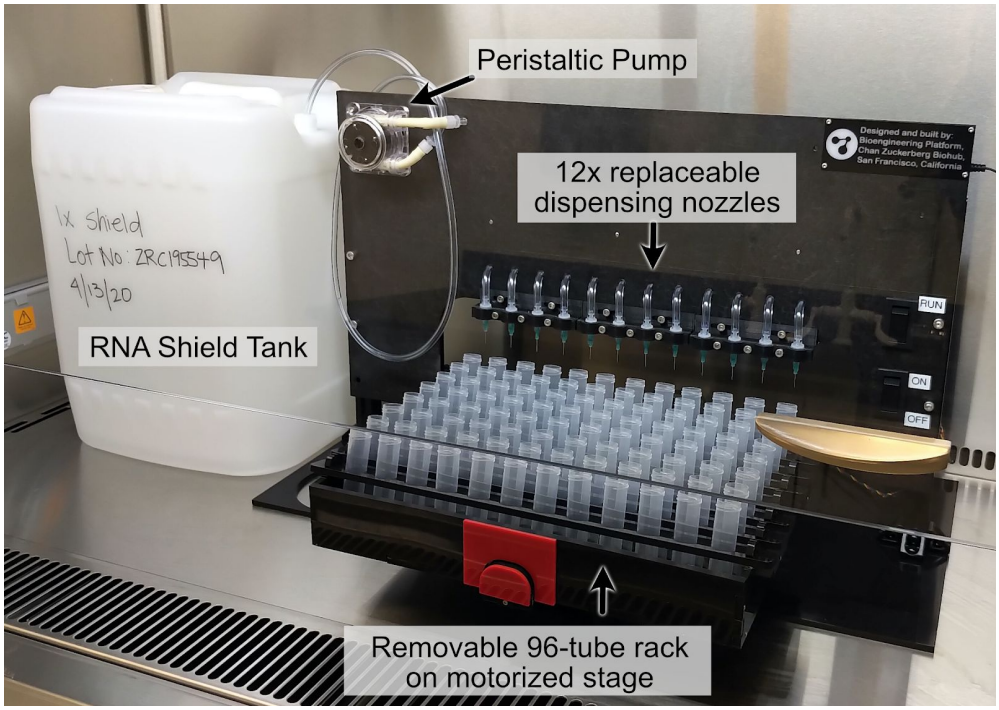


Figure XX. Bartender device developed to quickly fill tubes with RNA Shield. The device will automatically fill a rack of 96 tubes with a set volume of RNA Shield, with single button operation. The tube rack is removable from the machine, and multiple racks were built, so that some racks can be populated with uncapped tubes, while one rack is filled and others are

capped after filling. To preserve the cleanliness of the tubes and the Shield, the device is always operated inside a biosafety hood.

E. Collection Tube to Plate Transfers

Extensive use of liquid handling automation is a key factor in achieving a high testing throughput and reducing potential errors. The Automation team was tasked with exploring automation solutions to eliminate workflow bottlenecks and sources of error.

Manual Sample Tube to Multi-Well Plate Transfers - Well-Lit

Patient samples arrive at the UCSF Clinical Microbiology lab as individual collection tubes. To achieve greater throughput, patient samples must be accessioned and then transferred from each tube to a designated well on a barcoded 96-well plate. Transferring into a 96-well plate by hand pipet is a very slow and error-prone process. To minimize the chance of error, the manual process for the tube-to-well transfer involved two people, one doing the pipetting and the other registering the tube barcodes and writing down which well received each sample. Multiple steps of cross-checking between the two operators to avoid errors were necessary during the transfer, which increased the time needed to fill one 96-well plate to at least 2 hours.

Several automated solutions for tube-to-plate accessioning exist, and are discussed below. In the context of this particular pandemic emergency, these automated solutions were not readily accessible or sufficiently validated for clinical deployment. To immediately address the shortcomings of manual transfer, the Biohub Bioengineering team executed a rapid device design and build. The device design was necessarily based almost exclusively on parts already in-hand. The device is composed of a 12.3” computer touch-screen (scavenged from an older project), a miniature Windows PC computer (removed from a screen used to display conference room occupancy), a barcode scanner, and various custom-designed parts that are laser-cut and 3D-printed in-house.

We named this device “Well-Lit” because of the way it functions. The 96-well plate (2mL deep-well) that receives the samples rests on top of the touch-screen, held in place by a custom-made holder, and a pattern of brightly colored discs is displayed on the screen underneath the plate, one disc per well, so that the wells are illuminated with different colors to indicate to the operator which wells already have samples, and which well should receive the next sample, as the user scans each tube with the barcode scanner (Figure XX). Use of the “Well-Lit” device reduces the chance of error by providing clear visual guides for each sample, while simultaneously tracking and checking input barcodes. The system ultimately exports a LIMS-compatible file describing the operator, plate barcode and sample well positions, along with their respective barcodes.

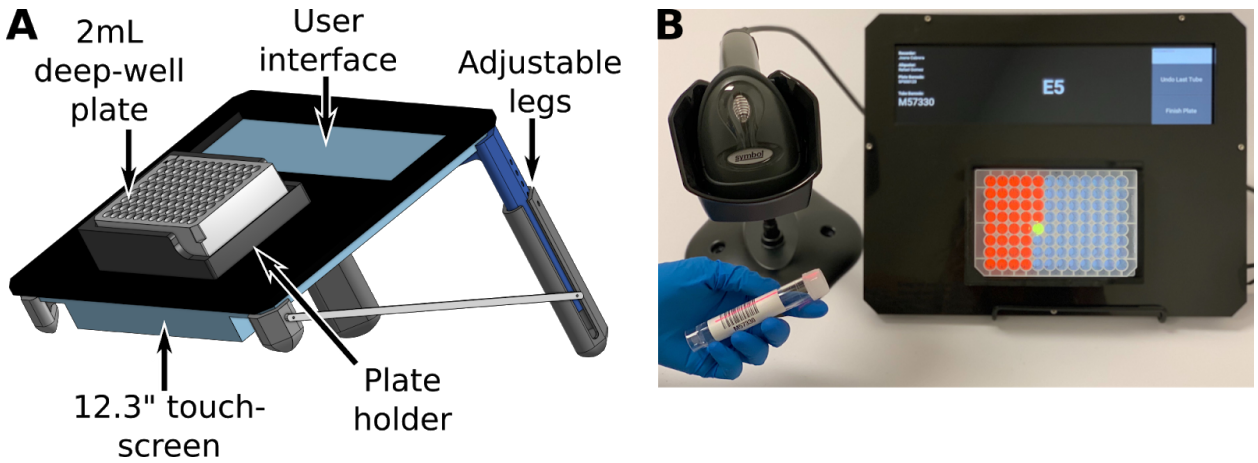


Figure XX. Well-Lit device developed to expedite the manual tube-to-well pipetting process, while simultaneously minimizing the chance for error. A. CAD rendering showing the basic structure of the device: 3D printed legs hold a 12.3” touch-screen at an angle to allow the operator to see straight into the wells in the plate (the legs are designed to allow adjusting the angle between approximately 30° and 46° with respect to the table, to accommodate different operator heights). A plastic panel with two laser-cut openings, one for the multi-well plate, and the other for a user interface area, is placed over the screen. A 3D printed plate holder bolted to the panel provides a key so that the plate can only be inserted in one orientation, and also shades the sides of the plate from ambient light to improve the contrast of the color discs illuminating the wells. The front panel is designed to be easy to place and remove for decontamination, and to allow panels designed for different types of multi-well plates to be easily swapped by the operator. B. The Well-Lit in operation, with the barcode scanner used to scan the barcodes on the multi-well plate and patient sample tubes. Wells that have already received samples are illuminated in red, while the well that should receive the sample from the tube that was just scanned is illuminated in yellow. The user interface area shows the operator name, plate barcode, sample barcode, and current well. Buttons on the right end of the interface allow the user to undo the last tube scan and finish the plate.

Automatic Sample Tube to Multi-Well Plate Transfers

To fully automate tube-to-plate transfer, the automation team elected to purchase the Hamilton STARlet liquid handling robot. The Autoload option on the STARlet automatically scans barcodes on sample tubes and destination plates and outputs a file used to assign results to samples. The Apricot A4, an alternative but lower throughput system that also automates the tube-to-plate transfer and records barcode information, may also be used. The A4 has the advantage of a smaller footprint, which allows it to fit into an extended biosafety cabinet. The STARlet was not inside a biosafety cabinet and thus was only used for processing samples already in DNA/RNA Shield; samples that arrived in UTM or other non-inactivating media were plated using Well-Lit inside a biosafety cabinet.

The Hamilton STARlet system was configured to automatically re-array 12 controls and up to 84 samples from tubes into a deep well plate. This critical step (or its equivalent on the Well-Lit

system) is the first point in the pipeline where the order of the samples is established. The deep well plate locations assigned during this step are maintained throughout the rest of the diagnostic workflow.

At the beginning of each run, the user has the option to change a variety of parameters in an interactive GUI. The fields that can be adjusted are: the number of patient samples, the path the pipette tips take from the tube to the deep well plate, the number of deep well plates aliquoted into, and the volume of each transfer.

The deck layout and path of the pipette tips from the tubes to the deep well plate minimize runtime and cross-contamination (**Supplemental Figure XX**). Another feature used to minimize cross-contamination was capacitive liquid level detecting to track liquid levels while aspirating or dispensing samples. The pipette will always aspirate or dispense while following the liquid level to minimize the surface area of the tip covered by the viscous sample. This will decrease the likelihood of a droplet of liquid forming on the outside of the pipette tip, which could potentially contaminate an adjacent well.

Before loading the samples onto the STARlet, users must invert and centrifuge the sample tubes to ensure that they are fully deactivated with Shield and that no bubbles are present. Users will then remove swabs from the samples then load them onto the STARlet. The STARlet will automatically load the samples, tips, and plates while simultaneously reading their barcodes. Once everything is successfully loaded, the STARlet will begin the rearraying protocol which takes approximately 20 minutes for a single plate transfer. A csv file is generated at the end of a run that contains the plate barcode, sample tube barcode and associated deep well plate ID. This csv is then submitted to the Biohub LIMS system for downstream processing.

F. Reagent preparation

All reagents required for sample preparation, RNA extraction and qPCR analysis were prepared in advance in a positive pressure, RNAse-free, pre-PCR environment to avoid possible sample degradation and contamination. All reagents used for RNA isolation prior to qPCR were part of the Quick-DNA/RNA Viral MagBead kit (Zymo Research). For each reagent, we used a specific type of 96-well plate, labelled in a way unique to each reagent in order to avoid confusion (Table XX). All plates were barcoded (Table XX) and logged in the laboratory reagent tracking system immediately after their preparation, along with information on date of preparation, batch of all reagents and the person that prepared the plate. For the preparation of reagent plates in the most time efficient and high throughput manner we used different types of liquid transfer automation (Table XX) depending on the type and volume of each reagent. We made use of three different liquid dispensers, namely the Multidrop™ Combi Reagent Dispenser (ThermoFisher), the Viaflo 96 well microplate dispenser (Integra) and the DragonFly Discovery (TTP Labtech). All reagents dispensed using a Multidrop™ (Thermo Fisher, cat. 5840300) were dispensed using reagent specific cassettes to avoid reagent cross-contamination.

In the following section we provide more detailed information regarding the preparation of each reagent.

Plate	Barcode tempate	Volume Needed Per Well (uL)	Instrument used	Plate Type	Storage Temperature
Proteinase K	RPPKXXXXXX	3	Integra Viaflo	1mL Deep Well	-20C
Viral DNA/RNA Buffer	RPVBXXXXXX	450	Multidrop™	1mL Deep Well	RT (Light-sensitive)
MagBinding Beads	RPMBXXXXXX	30	Integra Viaflo	96-well PCR plate	RT
MagBead Wash Buffers 1	RPWB1XXXXXX	550	Multidrop™	1mL Deep Well	RT
MagBead Wash Buffers 2	RPWB2XXXXXX	550	Multidrop™	1mL Deep Well	RT
Ethanol Wash Buffer 1/2	RPET12XXXXXX	1100	Multidrop™	1mL Deep Well	RT
Ethanol Wash Buffer 3/4	RPET34XXXXXX	1100	Multidrop™	1mL Deep Well	RT
DNase I	RPDNXXXXXX	55	Multidrop™	96-well PCR plate	4C
DNA/RNA Prep Buffer	RPRPXXXXXX	550	Multidrop™	1mL Deep Well	RT
DNAse/RNAse Free Water	RPH2OXXXXXX	100	Multidrop™	96-well PCR plate	RT
Sample plates (2X DNA/RNA Shield)	SPXXXXXX	500	Multidrop™	2mL Deep Well	RT
qPCR MM and primer plates	Random 8 digit	6	dragonfly discovery	384-well qPCR plate	-20C

Table XX. Summary of reagent information.

Sample plates (DNA/RNA Shield)

Sample plates containing 500 uL DNA/RNA Shield were used when transferring patient swab samples from individual tubes of UTM, VTM or other non-inactivating media to 96-well plates with Well-Lit. Each plate (Table XX) was filled with 500ul of 2X DNA/RNA Shield™ using a Multidrop™. Plates were sealed using a SuperSealer (Thermo Fisher, cat. 4110-11) and 96-Well cap Mat seals (Thermo Fisher, cat.12-565-556). After the plates were sealed, they were barcoded (SPXXXXXX) on at least two sides for easier identification, logged in the reagent tracking system and stored at RT until used.

Viral DNA/RNA Buffer

Before aliquoting, 2-mercaptoethanol was added to the Viral DNA/RNA buffer at a ratio of 500ul 2-mercaptoethanol per 100mls buffer. After mixing, multiple bottles of Viral Buffer were combined in sterile 1L bottles prior to aliquoting. Buffer was then aliquoted into 96 well deep well plates (Fisher brand, cat.12566611) at 450ul per well using the Multidrop Combi (cat. 5840300) and standard cassette (cat. 24072670) from ThermoFisher. After dispensing, plates were heat sealed using an ALPS 50-V sealer and Thermo Easy Pierce foil sheets (cat. AB-0757). All plates were barcoded (RPVBXXXXXX), logged according to protocol, and labeled “Viral DNA/RNA Buffer”. Aliquoted, sealed plates were stored at room temperature, in a flammable cabinet, protected from light.

DNAse/RNAse free water

Nuclease-free water used to elute RNA from the MagBinding beads was aliquoted in skirted 96-well PCR plates (Bio-Rad, cat. HSP-9641) using 100ul of water in each well. Water was dispensed using a Multidrop™. Following dispense, plates were manually sealed using foil PCR plate seals (Biorad, cat. MSF1001), barcoded (RPH2OXXXXXX) and labelled, logged in the reagent tracking system and stored at RT until used.

MagBeads

MagBeads are among the trickiest reagents to dispense given their viscosity and quick sedimentation rate. We ended up using the Viaflo 96-well microplate dispenser to transfer 30ul of MagBinding Beads from a deep well source plate to skirted 96-well PCR plates in the following way. The stock solution of MagBinding Beads was resuspended into a 25mL reagent reservoir (VWR, cat. 41428-958). 350uL of beads were transferred using an 8-channel pipette into deep well 96 well plates (Thermo Fisher, cat. 12-566-612), making sure to mix beads in the source reservoir every two or three columns. From the same reservoir, pipette another 450ul of MagBinding beads in each well of the same source plate for a total volume of 800uL. The plate was then mixed using a Viaflo, by aspirating and dispensing 300ul for a total of 20 times. Before the beads were allowed to settle, the Viaflo was used to dispense 30uL of beads from the source plate to the destination PCR plate (Biorad, cat. HSP-9641). Following dispense, plates were manually sealed using foil PCR plate seals (Biorad, cat. MSF1001), barcoded (RPMBXXXXXX) and labelled, logged in the reagent tracking system and stored **inverted** at RT until used. We found that when tubes were not stored in an inverted manner, beads settled to the bottom of the conical wells and did not resuspend well during extraction. When stored inverted, surface tension prevented the bead solution from dripping onto the plate seal and the beads resuspended easily. A total of 14 plates were produced per round.

MagBead Wash Buffers 1 and 2

MagBead Wash Buffer 1: Magbead Wash Buffer 1 was diluted with 100% isopropanol as directed by the manufacturer. 80mls of isopropanol was added to 120mls of concentrated buffer and mixed thoroughly. Buffer was aliquoted into 96 well 1ml deep well plates (Thermo Fisher, cat. 12-566-612) at 550ul per well using the Multidrop Combi and standard cassette (cat. 24072670) from ThermoFisher. After dispensing, plates were heat sealed using an ALPS 50-V sealer and Thermo Easy Pierce foil sheets (cat. AB-0757). All plates were barcoded (RPWB1XXXXXX), logged according to protocol, and labeled “Wash Buffer 1”. Aliquoted, sealed plates were stored at room temperature, in a flammable cabinet, protected from light.

MagBead Wash Buffer 2: Magbead Wash Buffer 2 was diluted with 100% isopropanol as directed by the manufacturer. 120mls of isopropanol was added to 80mls of concentrated buffer and mixed thoroughly. Buffer was aliquoted into 96 well 1ml deep well plates (Thermo Fisher, cat. 12-566-612) at 550ul per well using the Multidrop Combi and standard cassette (cat. 24072670) from ThermoFisher. After dispensing, plates were heat sealed using an ALPS 50-V sealer and Thermo Easy Pierce foil sheets (cat. AB-0757). All plates were barcoded (RPWB2XXXXXX), logged according to protocol, and labeled “Wash Buffer 2”. Aliquoted, sealed plates were stored at room temperature, in a flammable liquid cabinet, protected from light.

Ethanol Wash buffers

100% Ethanol was aliquoted into 96 well 1ml deep well plates (Thermo Fisher, cat. 12-566-612) at 1100ul per well using the Multidrop Combi and standard cassette (cat. 24072670) from ThermoFisher. After dispensing, plates were heat sealed using ALPS 50-V sealer and Thermo Easy Pierce foil sheets (cat. AB-0757). All plates were barcoded (RPET12XXXXXX or RPET34XXXXXX), logged according to protocol, and labeled “Ethanol Wash 1 and 2” or “Ethanol Wash 3 and 4”. Aliquoted, sealed plates were stored at room temperature in a flammable liquid cabinet.

Proteinase K

We used the Viaflo 96 well microplate dispenser to dispense 3ul of resuspended Proteinase K from a skirted 96 well PCR source plate (Biorad, cat. HSP-9641) to a 1ml deep well 96-well destination plate (Thermo Fisher, cat. 12-566-612). Proteinase K plates are stable for at least one week stored at -20C. Approximately 40 plates of Proteinase K were prepared as follows. 12 tubes of lyophilized Proteinase K (20mg) were spun down, then 1mL of Zymo Proteinase K Storage Buffer was added to each tube, resuspended gently, and then transferred into a 15mL Falcon tube. Into the same Falcon tube, add an additional 2.4mL of Proteinase K Storage Buffer and mix gently by inverting. Final volume was 14.4mL of resuspended Proteinase K. The contents were then poured into a 25mL reagent reservoir (VWR, cat. 41428-958), and then 130uL of Proteins K was transferred using a 12-channel pipette into the wells of a skirted 96 well PCR plate. The plate was subsequently spun down at 1000xg for 1 minute. Finally, the Viaflow was then used to transfer 3uL of Proteinase K from the source plate to the destination plate. Following dispense, plates were sealed using foil PCR plate seals (Biorad, cat. MSF1001), barcoded (RPPKXXXXXX) and labelled, logged in the reagent tracking system and stored at -20C until use.

DNAse I

We used the Multidrop™ to dispense 55ul of a solution of DNAse I in skirted 96-well PCR plates (Biorad, cat. HSP-9641). DNAseI plates are stable for at least one week when stored at 4C. Approximately 35 plates were prepared per round as follows. 12 tubes of lyophilized DNase I (1500U) were spun down, and then resuspended with 1mL of nuclease-free water each, followed by transfer to a large reservoir (200mL). In the same reservoir, 166mL of nuclease-free water was added and gently mixed, followed by addition of 19.8mL of DNA Digestion Buffer. 55uL of the solution was dispensed to each well of a skirted 96-well PCR plate (Biorad, cat. HSP-9641) using a Multidrop™. Following dispense, plates were manually sealed (Biorad, cat. MSF1001), barcoded (RPDNXXXXXX) and labelled, logged in the reagent tracking system and stored at 4C until used for up to a week.

RNA Prep Buffer

RNA Prep Buffer was aliquoted into 96 well 1ml deep well plates (Thermo Fisher, cat. 12-566-612) at 550ul per well using the Multidrop Combi and standard cassette (24072670) from ThermoFisher. After dispensing, plates were heat sealed using ALPS 50-V sealer and Thermo Easy Pierce foil sheets (cat. AB-0757). All plates were barcoded (RPRPXXXXXX), logged according to protocol, and labeled “RNA Prep Buffer”. Aliquoted, sealed plates were stored at room temperature, in a flammable liquid cabinet, protected from light.

qPCR master mix plates

The qPCR Master Mixes were prepared on ice according to the primer and probe concentrations that are listed in Table X (see section G). For the master mix, the Luna Universal Probe One-Step RT-qPCR Kit (NEB, E3006) was used. All mixes were formulated in an RNAse free, pre-PCR environment to avoid sample degradation and contamination. Each target master mix was assigned a quadrant within the four quadrant layout of a 384-well qPCR plate, and empty quadrants were filled with water. Six microliters of master mix or water were dispensed into each well of a barcorded 384-well qPCR plates using the DragonFly Liquid Dispenser (TTP LabTech), as described in Supplementary File X (see end of document). Plates were then sealed using Microseal “F” PCR Plate Seals (Bio-Rad, MSF1001) and stored at -20°C until use. When RNA extractions were completed, 4 µl of RNA sample was added to the 6 µl of master mix, resulting in qPCR reactions with a final volume of 10 µl.

G. RNA Extraction

Incoming 96 well 2mL deep well plates, sealed with single-use sealing mats (Thermo Fisher AB-0674), containing 1x Shield were kept at 4C and then queued for processing. Using the Agilent Bravo automated liquid handling platform, we were able to implement a high throughput extraction protocol adapted for the Zymo Quick-DNA/RNA Viral MagBead kit (R2141) because we were familiar with its high quality and consistency, it was cost effective, and it was not limited by supply chain shortages experienced by other vendors. Briefly, the Bravo RNA extraction protocol consists of: 1) 180uL sample transfer from 2mL deep well to a 1mL deep well plate containing Proteinase K; 2) Addition of Zymo Viral DNA/RNA Buffer for sample lysis; 3) Addition of Zymo MagBeads; 4) 10 minute mixing and shaking of samples with lysis buffer and magbeads; 5) incubation of the mixture on a 96 well ring magnet to collect the beads to a ring at the bottom of the deep well plate; 6) aspiration of the supernatant and dispensing into a 2mL deep well waste plate; 7) Addition of wash buffers 1 with mixing; 8) incubation on the 96 well ring magnet; 9) aspiration. Steps 7-9 are repeated for wash buffer 2 and two rounds of 100% ethanol. 10) Addition/mixing of DNAse mix and incubation for 15 minutes; 11) Addition/mixing of RNA prep buffer to deactivate/remove DNAse; 12) repeat of steps 7-9 with 2 rounds of 100%

ethanol; 13) incubation on the magnet for 20 minutes to fully evaporate residual 100% ethanol from the beads; 14) Elution with nuclease-free water. Eluted RNA is aliquoted into pre-made qPCR plates that are frozen at -20C until ready to use. Leftover RNA is frozen at -80C.

A detailed protocol, including Agilent Bravo Program Files, is available on protocols.io.

H. RT-PCR assay

Assay design

The WHO published several RT-PCR assays that were established shortly after the spread of Sars-CoV-2 by the national laboratories of different countries. Before setting up the Covid-19 task force at CZ Biohub, anecdotal reports on suboptimal performance of the CDC assays had emerged. We compared the performance of the German E and RdRp assays, and the Japanese, Chinese and Hong Kong N gene assays using RNA extracted from viral cultures, received from the CDC, at an extension temperature of 58°C using the ThermoFisher TaqPath 1-step RT-qPCR Mastermix. Alignment of the Japanese reverse N gene primer revealed a mismatch with the available genomes in GISAID. One base of this primer was therefore modified to ensure 100% identity with the available sequences. These primers were rush-ordered at small quantities for initial testing of the qPCR assay as quickly as possible.

Target	**Primer/probe name**	**Primer/probe sequence**	**Final concentration**
E (viral, Germany)	SARS-CoV-2_E_Fwd	ACAGGTACGTTAATAGTTAATAGCGT	400 nM
	SARS-CoV-2_E_Rev	ATATTGCAGCAGTACGCACACA	400 nM
	SARS-CoV-2_E_Prb	ACACTAGCCATCCTTACTGCGCTTCG	200 nM
N (viral, Japan)	NIID_2019-nCOV_N_F2	AAATTTTGGGGACCAGGAAC	500 nM
	NIID_2019-nCOV_N_R2	TGGCACCTGTGTAGGTCAAC	500 nM
	NIID_2019-nCOV_N_P2	ATGTCGCGCATTGGCATGGA	200 nM
RNAseP (host)	RNAseP Forward	AGATTTGGACCTGCGAGCG	200 nM
	RNAseP Reverse	GAGCGGCTGTCTCCACAAGT	200 nM
	RNAseP P	TTCTGACCTGAAGGCTCTGCGCG	50 nM

Table X. Primer and probes used to detect SARS-Cov-2 using RT-qPCR. The E and N probes were labelled with 5' 6-FAM™/ZEN™/3' IB®FQ, and the RNAseP probe with 5' HEX™/ZEN™/3' IB®FQ. All reactions were performed in separate wells. The cycling protocol is described in Supplementary File X.

When the different assays were compared using their published optimal primer concentrations at an extension temperature 58°C, the E and N gene assays proved to be the most sensitive (lowest Ct values), with the exception of the HK N-gene assay (Figure SX). The modified

Japanese primer set performed best among the N gene assays. The water and HeLa RNA controls returned no Ct values, as expected.

After these initial tests, we placed orders for large-scale synthesis of primers and probes for the E gene and the modified Japanese N gene assays, and the RNAseP assay from Integrated DNA Technologies (IDT). The RNAseP probe was labelled with a HEX fluorophore and the E gene probe was ordered with FAM. Importantly, these primers and probes were produced in template-free facilities to minimize the chance for contamination as much as possible.

The RT-PCR assay consisted of three single reactions using the NEB Universal Probe One-Step RT-qPCR kit, all performed in the same 384-well qPCR plate. The NEB kit was preferred since it could be more easily sourced than the widely used ThermoFisher TaqPath 1-step RT-qPCR Mastermix. qPCR plates containing 6 µl mastermix were prepared using the Dragonfly dispensing robot (SPT Labtech), with quadrant A1, A2 and B1 containing the RdRP, E and RNAseP mastermixes, respectively. Quadrant B2 was filled with water. When RNA extractions were completed, qPCR plates stored at -20°C were thawed and 4 µl of RNA was added to the 384-well qPCR plate using the Bravo robots. This format reduced the chance of error by eliminating any re-arraying; each well of the 96-well RNA plate was stamped into three of the four corresponding wells on the 384-well plate.

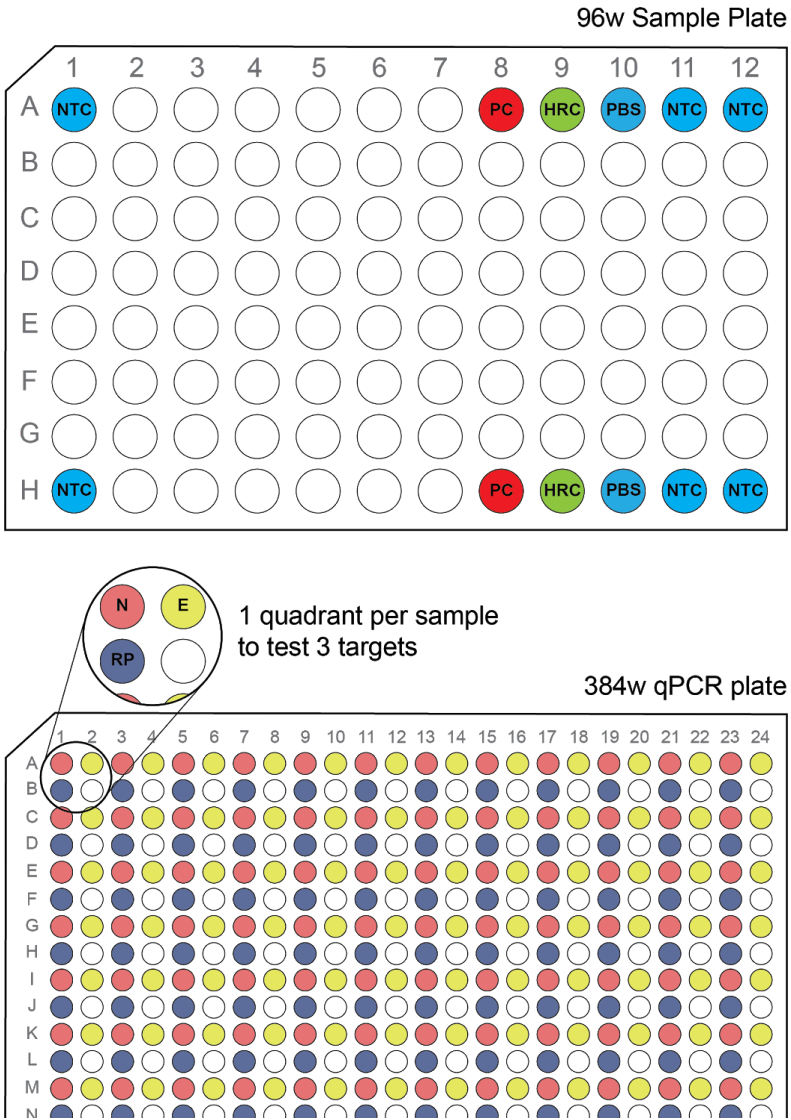


Figure X. Layout of the sample and qPCR plates. The 96w sample plate has 12 controls that are asymmetrically plated so that accidental rotation of the sample or qPCR plate at any point during the extraction process can be detected. After extraction, 4 µl of RNA is printed into 3 of the 4 wells in the corresponding quadrant of the 384w qPCR plate to allow detection of the N, E and/or RNaseP (RP) genes. NTC: no template water control, PC: positive control, HRC: human RNA extraction control.

Assay controls

We included four types of controls on the 96-well sample plate: the no template water control (NTC), the negative extraction control (PBS), human RNA extraction control (HRC) and the positive control (PC). In order to avoid manual pipetting steps during extraction or qPCR reaction setup, the no-template water control (NTC) for the qPCR reaction was plated out together with all samples before RNA extraction and essentially served as an extra negative extraction control. The HRC consisted of PBS:Shield with 100 ng/ml HeLa RNA, which resulted in an RNAseP Ct value of 30-31. In version 2 of the protocol, upon which all of the data in this manuscript is based, the PC consisted of HRC that was spiked-in with high titer patient sample to yield E- and N-gene Ct values of about 30. In version 3, developed later, the high titer patient sample in the PC was replaced by an in vitro transcribed (IVT) synthetic RNA construct that contained the different viral targets detected in our study in one single 600 nt RNA fragment (Supplementary File XX). SARS-CoV2 genomic regions containing the primer/probe target sequences for RdRp, E and N genes were PCR amplified from cDNA from a SARS-CoV2 positive patient sample. RdRp, E and N fragments were assembled using overlap extension PCR, cloned into pCR-BluntII-TOPO and a colony containing the insert in sense orientation relative to the T7 promoter was selected. The plasmid was linearized using BamHI (cut site situated downstream of the insert), RNA was generated from linearized plasmids using T7 RNA polymerase, treated with TurboDNase I and purified using lithium chloride. To quantify copies of IVT RNA, 10-fold serial dilutions of the IVT RNA were made and quantified by digital droplet PCR. Bio-Rad 1-Step RT-ddPCR Advanced Kit and the Bio-Rad Automated Droplet Digital PCR system were used. Our final IVT RNA stock is approximately 10^13 copies/mL. Serial 0.5log dilutions of the IVT RNA PC template starting at 10^7 copies/mL were tested to determine the final working stock for our positive control. A concentration of 10^4 copies of IVT RNA/ml was eventually selected since this consistently resulted in Ct values of 29-30 for the N and E gene assays.

Control	**Content**	**Expected result**		
		RNAseP	**N**	**E**
NTC	**H_2O:Shield**	No Ct value	No Ct value	No Ct value

PBS	PBS:Shield	No Ct value	No Ct value	No Ct value
HRC	**HeLa RNA in PBS:Shield**	Ct 30	No Ct value	No Ct value
PC	**HRC + HTP or IVT RNA**	Ct 30	Ct 29-31	Ct 29-31

Table X. Controls used during RNA extraction and RT-PCR. NTC: no template control, HRC: human RNA extraction control, PC: positive control, IVT: in vitro transcribed.

Assay interpretation

A clinical sample was considered positive ('Detected') for SARS-CoV-2 if both viral genes were detected with a Ct <40 (Table X). If only one viral target yielded a Ct <40 and the other was negative or yielded a Ct ≥40, the result for that sample was deemed "indeterminate" and the sample was re-extracted. If the same result would be obtained after re-extraction, the sample would be reported as indeterminate, and treated clinically as a presumed positive.

A clinical sample was considered negative ('Not detected') only if both viral targets were negative and a Ct <38 was observed for RNAseP. Negative samples with no RNAse P signal or RNAseP Ct values ≥ 38 were re-extracted. If both viral targets were negative after re-extraction and the RNAseP Ct was <40, the sample would be reported as negative or 'Not detected'. If the RNAseP signal was absent or RNAse P Ct value was ≥40 after re-extraction, we concluded there was not enough human material in the sample and requested that the patient be re-sampled.

N (viral)	E (viral)	RNaseP (host)	Interpretation	Action
Pos	Pos	Pos or Neg	Detected	Report as Detected
1 or 2 pos, or 1 pos with Ct value ≥40		Pos or Neg	Indeterminate*	Repeat with new extraction; if same result report as Indeterminate
Neg	Neg	Pos	Not detected	Report as Not Detected
Neg	Neg	Neg	Invalid	Repeat with new extraction; if same result report as Invalid.

Table X. Clinical result based on RT-qPCR results. Pos: positive, Neg: negative, Ct: threshold cycle. * in vB, this category was considered positive.

I. Data Flow, Analysis & Resulting: Building a lightweight, serverless LIMS

A good Laboratory Information Management System (LIMS) is essential for the smooth functioning of a diagnostic lab and is a requirement for CLIA validation. We needed a LIMS solution that would be quick to stand up and fast to customize, deliver reliable test results and satisfy CLIA requirements even as the wetlab configuration, personal, and processes were changing daily. We identified the following set of key requirements for a LIMS in a rapidly built COVID testing facility:

1. Ensure final results are accurate,
2. Efficiently report results back to our clinical partners,
3. Perform validation and on-going QC of laboratory assays,
4. Store auditable and redundant logs of all actions performed in the lab,
5. Comply with all regulatory requirements for storing Personally Identifiable Information (PII) and Protected Health Information (PHI),
6. Robustly scale to a large number of daily tests, and
7. Track samples, reagents and waste throughout the system, to enable rapid debugging, predict inventory needs and comply with regulatory requirements for waste disposal.

We first considered adopting a full-featured commercial LIMS, such as Sunquest, the system used in the clinical labs at UCSF. However, we quickly realized that a commercial system, while purpose built for reliability and regulatory compliance, would not satisfy some important needs of our rapidly built clinical testing laboratory. In particular, we were concerned with:

1. Time - commercial systems are slow to requisition and purchase and require vendor technicians to come on site for installation, leading to initial setup time of weeks or months, instead of days, and
2. Flexibility - the FDA authorized RT-PCR laboratory developed tests rapidly developed in response to the COVID-19 pandemic have unique data processing and logging

Upon evaluating how quickly we could make it work (time), how easy it would be to change (flexibility), how simply we could identify errors (transparency) and cost, we decided to build a custom based solution.

The first major decision was that our internal LIMS system would have no interaction with PII or PHI. All patient information is stored in the FDA-approved Sunquest system at the UCSF Clinical Laboratory. At the point of patient data entry into Sunquest, a semi-unique Accession Number is given to each sample (this consists of 5 or 6 characters and can repeat on the timescale of months). The corresponding tube is barcoded with this identifier, and it is read into the internal CLIAHUB LIMS during plating. The internal LIMS later matches the result to the accession number, and that data is returned to the UCSF Clinical Laboratory where it is entered into Sunquest using approved protocols and thus re-connected with patient data.

H.1. Lightweight, serverless LIMS

The CLIAHUB LIMS (Fig. DS1), our solution, is built on well-supported standard software tools:

- Google forms and sheets to capture actions in the lab
- Google Drive to store machine output, result files, and internal process/status
- AWS Lambda running python code
- Github for software development
- Slackbots for inventory alerts (low on reagents) and errors (missing metadata for a plate)
- Email for sending reports to shift supervisors and to China Basin
- Rich PDF reports, with QC information as well as test results

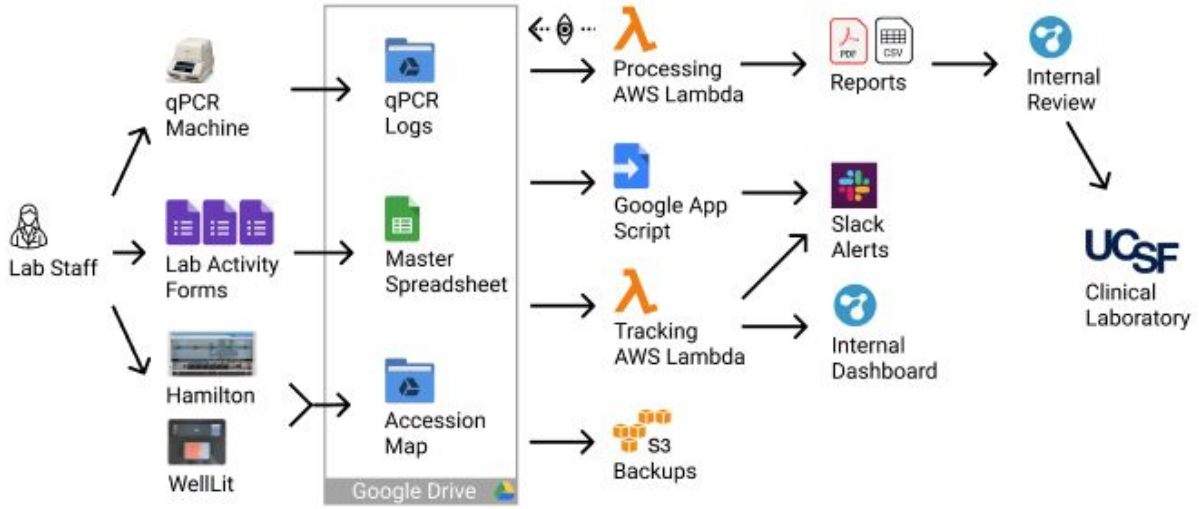


Figure DS1. Architecture of a lightweight, serverless LIMS. Sample processing data is captured manually by lab personnel in Google forms and qPCR results automatically sync to Google Drive. An AWS Lambda process spins up and polls the Drive every 4 minutes (dashed line) to monitor new files. If data from a new qPCR run is detected, a processing script parses the log files and generates pdf and csv reports, which are emailed to shift supervisors. The pdf is reviewed internally, and then both reports are sent to the main UCSF clinical lab for a final review before uploading into patients' medical records. Errors in data entry along the process are detected by a Google App Scripts script triggered on form submission, and alerts are sent to the lab personnel by a bot in Slack. Periodic backups of all data in Google Drive are made to AWS S3, another cloud-based storage system. Sample status can be tracked in an internal dashboard, which is periodically populated by a second AWS Lambda job. The entire data flow

can be reconfigured by modifying the forms, the contents of Google Drive, or the python scripts run by the AWS Lambda.

H.2. Implementation details

1. H.2.1. **Sample Flow**

One of the determinant steps was to break the workflow into stations and understand what information was relevant at each stage.

Sample prep: Samples are aliquoted from individual tubes into 96-well plates using the WellLit software. This links accessions to plate wells.

- **Sample Metadata**
    - Registers the plate metadata for new samples

Sample plate arrival: 96 Sample Plates are brought to CLIAHUB.

- **Sample Arrival**
    - Registers new samples arriving from China Basin
- **4C Plate Check In**
    - Registers new plates being stored at 4C

Bravo: First RNA is extracted, and each sample well on the 96 Sample Plate corresponds to a well in the 96 RNA Plate. Then each well of the 96 RNA Plate is aliquoted into 4 wells of the 384 PCR plate.

- **Bravo RNA extraction**
    - Logs the RNA extraction step and PCR Plate preparation, including all the reagent plates used

qPCR: Each well of the 384 is tested for one gene: RdRp (SARS-CoV-2), E (SARS-CoV-2), or RNAse-P (human gene). Update: or N (SARS-CoV-2) instead of RdRp.

- **PCR metadata**
    - Registers the metadata associated with each PCR run

Calls: Gene-calls from each well are linked back to initial accessions. Final calls (pos/neg/ind/inv) are made.

- qPCR machine output is automatically copied to google drive and processed (this step reduces error and allows users to leave the lab prior to completion of the 105-minute qPCR.

H.2.2. Data processing

In a nutshell, after the qPCR completes, t*he results for each gene are traced back to the initial sample accessions to produce final calls. A pdf report with barcodes is then generated [add figure with pdf, likely a supplementary figure to allow us to show a full report] and sent to clinical lab. While this fulfills the goal of the CLIAHUB data team is to deliver accurate, actionable, and auditable information about COVID19 cases to clinicians, public health professionals, and other scientists, t*he automated data processing is a fantastic piece of software engineering that glues the sample flow.

[add details on all the different lambdas we have, running times]

The results are compiled in our internal dashboard [add link + description]

J. Validation and Stability

On March 16th, the FDA issued the "Policy for Diagnostic Tests for Coronavirus Disease-2019 during the Public Health Emergency" [https://www.fda.gov/media/135659/download]. In this document, the FDA put forward several validation study recommendations for SARS-CoV-2 molecular diagnostic tests.

The first recommendation related to documenting the limit of detection (LOD) of the SARS-CoV-2 assay. We prepared LOD plates using a ddPCR-quantified high titer patient sample, provided by the clinical lab of UCSF, and diluted it to $10^{2.5}$, $10^{2.0}$, $10^{1.5}$ and $10^{1.0}$ viral genome copies per ml. Twenty replicates of each concentration were plated out together with 16 negative controls (total of 96 samples). This plate was run on all of the 8 stations used for RNA extractions and RT-PCR. The FDA defined the LOD as the lowest concentration at which 19 out of 20 samples are detected. In our case, we required an E gene or an N gene Ct < 40 in order to call a sample positive. The LOD of our assay was 10^2 viral genome copies per ml when sampling 160 µl for extraction and using 4 µl (out of 40 µl) eluted RNA for the qPCR reactions (Table X).

Concentration (viral copies/mL)	Total # of Reactions	# of Positive Reactions	# of Negative Reactions
317.0 ($10^{2.5}$)	160	160	0
100.0 ($10^{2.0}$)	160	158	2
31.7 ($10^{1.5}$)	160	110	50
10.0 ($10^{1.0}$)	160	48	112
0 (Neg ctrll)	128	0	128

Table X. Limit of detection of RT-PCR assay was determined to be 100.0 viral copies/ml.

The second FDA recommendation involved testing a minimum of 30 non-reactive specimens and 30 contrived reactive specimens. Of the contrived specimens, twenty should be spiked at a concentration of 1×-2× the LOD. The others can contain higher concentrations that are within the detection range of the assay. In addition to 30 confirmed negative patient samples, we received from UCSF 20 negative patient samples that were spiked with high titer patient sample at 2× our LOD, another 5 samples at 5× our LOD, and another 5 samples at 10× our LOD. RNA was extracted and the qPCR was performed as mentioned for the LOD plate. Our assay correctly identified all positive and negative samples in the validation plate when tested at multiple stations. It's important to note that the FDA requirements in this aspect changed along the way. Two months after publishing the Covid19 policy document and when reviewing previously submitted EUA submissions, authorities requested to test about 100 negative and 30 positive samples that were initially tested on an (non-EUA) approved platform for validation.

The third and fourth recommendations relate to primer and probe design. The sequences of primers and probes should be 100% identical to the available genomes of SARS-Cov-2 to ensure that all known variants can be identified. This was verified using the available genomes in GISAID and pubmed. We earlier identified one mismatch in the reverse primer of the Japanese N gene assay, and corrected this when ordering the primers. In addition, cross-reactivity with other pathogens should be minimal. The FDA defined that primer and probe sequences should have ≤80% homology to common respiratory commensal and viral pathogens.

In addition to the FDA guidelines, we also verified the level of cross-contamination that could potentially be caused by high titer samples during high-throughput RNA extractions. We prepared 3 plates that had each 6 random positive wells containing high titer sample. This high titer sample consisted of 10^9 copies/ml in vitro transcribed RNA containing the E and N gene fragments that are detected in our assay. This resulted in N-gene Ct values of about 12 and E-gene Ct values of about 11 in the positive wells. Each plate was run 3 times on different stations using the vB extraction protocol, and the number of positives (POS) by cluster and the number of POS in non-adjacent wells were recorded. Positivity was defined as a well that has an N- and/or E-gene signal.
For a total of 54 intended positives (6*3*3*), we observed 10 contamination events in adjacent wells, labelled as positives by cluster, and 2 contamination events that were 2-3 wells away from the closest high titer sample. The average number of contamination events per high titer sample was therefore found to be 0.22 (12 events for 54 high titer samples).
Based on these data we decided to routinely repeat positive clinical samples that (i) have a Ct difference >15 compared to an adjacent positive sample, and (ii) samples that have a Ct difference >22 compared to a positive sample within 3 wells.

	# POS by cluster	**# POS non-adjacent**

	(# - N Ct, E Ct)	(# - N Ct, E Ct)
Plate 1		
Station 6	1 - N:36, E:34	-
Station 5	1 - N:37, E:-	-
Station 4	1 - N:36, E:- (Same well as B130205)	-
Plate 2		
Station 7	-	-
Station 6	1 - N:38, E:35	-
Station 5	3 - N:35-36, E:34-36	1 - N: 36, E:36 (Located 2 wells away)
Plate 3		
Station 4	-	1- N: 34, E:34 (Located 3 wells away)
Station 8	2 - N:-, E:35-36	-
Station 7	1 - N:36, E:34	-

Table X. Cross-contamination caused by high titer samples is limited. Data shows the number of cross-contamination events and the range of N and E gene Ct values that were observed in these wells. As a reference, the high titer samples showed N and E gene Ct values of 12 and 11 respectively.

The different media used to collect samples, other than Shield, required validation before such samples could be accepted for processing. To validate a new media type, we first prepared a stock of concentrationd HeLA RNA in which a high titer patient sample was diluted to yield Ct values of 30 for the N, E and RNAseP assay. This stock was then diluted 1:4, 1:16, 1:64 and 1:256 in the media types to test. Three replicates of each concentration were plated out alongside the same dilutions in Shield as a control. These dilutions should result in Ct values of about 32, 34, 36 and 38. A medium type is validated for use if there is 100% detection for the 1:4 and 1:16 dilutions. Media types that were validated on our platform were Universal Transport Medium from COPAN, purple UTM, alternative UTM (Hanks + PBS + ampB + gentamycin), Viral Transport Medium from MANTACC, Viral Transport Medium from Abbott , and Amies Transport Medium. The Aptima medium was found to be incompatible with our platform since it precipitated when mixed with Shield.

To verify the performance of DNA/RNA Shield as a storage medium, we compared it to saline. Tubes with IVT RNA at a concentration of 10^3.5 copies per ml were prepared on day -16, d -8, d-4, d-3, d-2, d-1 and d 0, and stored either at room temperature or at 4°C until day 0. On d 0, all samples were aliquoted out on one single plate for extractions and qPCR. Results indicated

that Shield preserved IVT integrity for minimum 16 days when stored at room temperature or 4ºC. In saline, IVT RNA was undetectable after 8 days, independent of where it was stored. These data support that Shield is indeed the better storage medium.

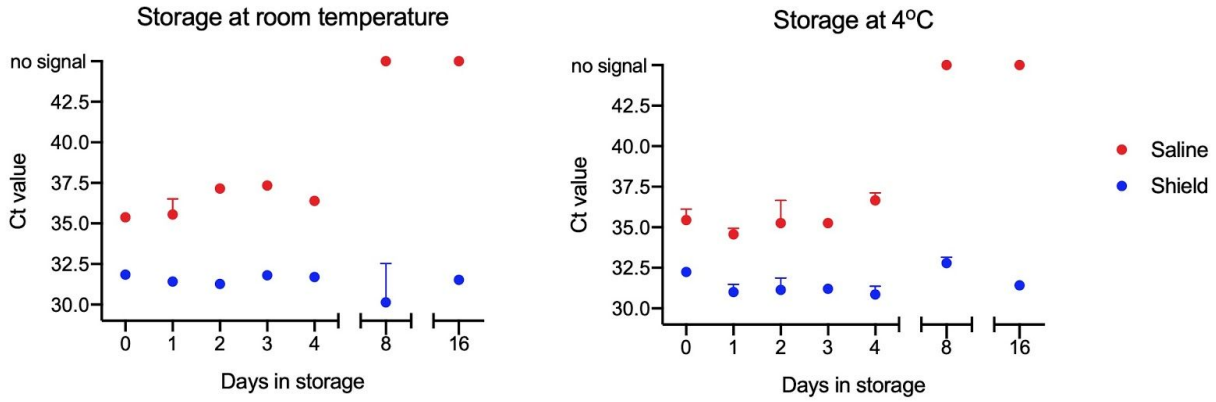


Figure X. Shield performs better than saline to store RNA at room temperature or at 4ºC. Experiment was performed using IVT RNA at a concentration of 3162 copies (10^3.5) per ml. All samples were processed on the same sample and qPCR plates to minimize technical variation. Error bars indicate the standard deviation based on 2 technical repeats of the qPCR assay.

K. Testing Data Summary

L. mNGS follow-on work

In coordination with the CLIAhub facility, a companion mNGS workflow was spun up to investigate the frequency of co-infections among COVID19+ patient specimens and enable real-time genomic epidemiology of circulating SARS-CoV-2 detected at the CLIAhub. Samples for these studies were acquired in parallel from internal and external sources. Internal samples were hit-picked from our onsite UCSF CLIA laboratory. External samples were acquired through collaborations with different research and CLIA laboratories in the Bay Area, as well as Departments of Public Heath (DPH) throughout North California. Sample tracking and processing for both these workflows was handled in parallel as described below.

Samples were received as extracted RNA and immediately arrayed into 96 well plates for storage, processing, and tracking purposes (Figure Xa). Internally sourced samples were cherry-picked into the inner 60 wells of 96 well working archive plates

(COMET96-RX-RXX). Restricting to plating to the inner 60 wells streamlined plate merging and minimized potential edge effects for downstream 384 well library prep.

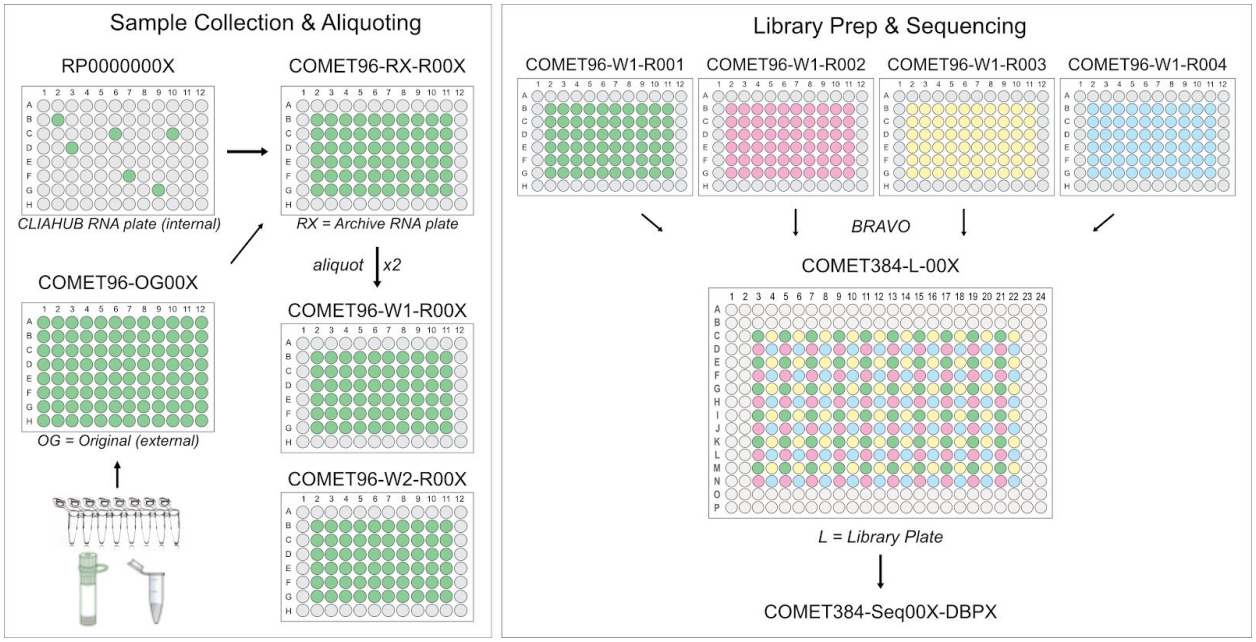
A LIMS system was applied to flag when 60 samples that fit a user-provided threshold of qPCR positive, indeterminant, and negative samples had accumulated during ongoing CLIAhub testing. The LIMS system captured relevant metadata, along with the RNA plate and well information for each sample. This facilitated hit-picking from CLIAhub source plates to destination working archive plates (COMET96-RX-RXX) for mNGS. Two working plates, each with 7ul volume per well were stamped from each working archive plate. In total, this workflow generated 3 replicate plates for: archiving (RX), library preparation (W1), and library preparation backup (W2).

Externally sourced samples, typically provided in tubes, were first transferred into a 96 well plate on receipt in the lab (COMET96-OGXXX, original plate). Resulting "OG plates" were used for long term storage of each sample and a source for hit-picking samples of interest for library preparation. Limited metadata, (e.g., sample identifiers, sample volume, COVID19 qPCR assay calls and C_t values, name of associated lab or testing site, etc.) associated with each sample were recorded in a standard spreadsheet template, which was registered into the mNGS LIMS through an accompanying Google form. Each sample was also assigned a unique identifier (CZB_ID).

Generation of a working archive plate for external samples followed OG plate generation. The CZB_IDs of 60 samples of interest among any of the OG plates were compiled into a csv file. The csv file was uploaded to the mNGS LIMS, and the system returned the OG plate and well location of the sample. As described above for internal samples, cherry picking of OG plates also resulted in 3 replicate plates for: archiving, library preparation, and library preparation backup.

For 384 well library prep and sequencing, an Agilent BRAVO liquid handler was used to combine four individual working plates into one 384 well plate for miniaturization and high-throughput processing with the Beckman Echo Acoustic liquid handler (Figure Xb). Operators used a Google form to identify the four working plates as input and receive a 384-well plate map as output (COMET384-L-XXX). A spiked SARS-CoV-2 mNGS NEBNext Ultra II RNAseq library prep was applied to generate final sequencing libraries. External RNA Consortium Control Spike-In Mixes (ERCCs) were incorporated into each sample, and a dilution series of Hela RNA control samples were incorporated

into every sequencing run. Once the plate was multiplexed with dual barcodes, additional google forms in the LIMS were used to combine the 384 well plate map with the appropriate 384 well dual index barcode plate map to generate a sample sheet for the appropriate sequencer (COMET384-SeqXXX-DBPX). Libraries were pooled Paired-end 2x150bp sequencing was perfomed on the NovaSeq 6000, to an average read depth of 25M-50M reads per sample.



L. Personnel, Volunteer Management

Recruiting. Rapid expansion of operations for clinical testing puts an immediate stress on the pool of qualified professionals needed to run the tests, especially when those individuals are already in high demand and working long hours. However, when you are part of a University health care system that trains graduate students in the biomedical sciences, skilled labor is not a scarce commodity. With the sheltering in place orders in effect, most research laboratories were shut down and medical student rotations put on hold. This left a large cohort of hard working students without anything to do, and extremely eager to help in any way possible. As news spread of our clinical operations, we were soon flooded with offers to help and it became readily apparent that we needed a more formalized personnel team to organize, track and communicate with the volunteer workforce.

Job Training. As the technical pipeline was constructed, management of specialized job categories was accompanied by organizational onboarding, training and certification. In

compliance with CLIA regulations each volunteer that joined the operations was vetted for educational background, having to hold at least an Associates Degree in a related STEM field, and underwent specified training for each SOP within their job category. All documentation of onboarding and certification were kept as personnel files in accordance with UCSF Clinical Lab SOP.

Training of testing staff, consisted of a 3-day curriculum of on-the-job training for efficient integration into the workflow. Because the nature of HTP pipeline is not only learning to use liquid handling robots that implement the technical protocols, but also recognizing when things go wrong and understanding how to address issues. Thus, a hierarchical structure was implemented to maximize the capacity of this smaller group of more experienced scientists to oversee the newly trained volunteers.

As the workforce was trained up, squads were formed and included a supervisor, 2-3 technical experts and up to 20 testing technicians to run assays at one of the 8 robotics stations, accession samples, prepare reagent plates and dispose of waste. This approach allowed for rapid growth and onboarding of new volunteers as physical capacity for testing increased. Additionally, since testing needs were unpredictable for any given time of day, and dependent on supply chain dynamics for testing kits, working in squads provided flexibility to contract and expand the crew based on need.

Shift Work. Shift work was designed to serve multiple purposes, consistency of training, team bonding, easier schedule management and importantly, safety. Shifts were structured into early and late – 8 hour blocks per day, to maximize productivity without adding undue stress to the workforce. Each squad was assigned either an early or late timeslot and scheduled for 3 days on and 3 days off each week, with an alternating fourth day on, every other week. Setting up consistent work schedules set clear expectations around job responsibilities, and allowed individuals to achieve better time management and establish routines outside of the lab.

Training of each group under an individual supervisor allowed for smoother transfer of technical skills and knowledge necessary to be successful. Setting expectations within a particular squad helped to reduce stress in a high pressure work environment. Given that volunteers were gathered from across a variety of biomedical laboratories and departments, the formation of squads also allowed for team bonding, and created a familiar and more comfortable work environment. Squads facilitated comradery between workmates and boosted morale, and also allowed supervisors to pair individuals at different workstations effectively.

One of the biggest considerations for forming squads, was with regard to keeping volunteers healthy. Technical operations for HTP clinical testing needed a sizable workforce. Cohorting allowed each group limited exposure to a much smaller subset of the entire workforce. Together with restricted work hours, this approach limited overlapping between squads, and protected the larger group from potential exposures when an individual became ill. Cohorting also made it easier for supervisors to fill gaps when individuals were randomly unable to work.

Other Health and Safety.
The only real potential for viral exposure from sample handling, was at the main clinical lab where swabs were initially sent for processing. Thus, in addition to the squads, a separate cohort was created to work at the main lab site, and did not have any overlapping duties with

the others. This group of individuals were trained to pipet samples from tubes to plates, and then package them for transfer to the CLIAHub.

Answering a health questionnaire was required upon arrival to the lab everyday. Self-monitoring was recommended for tracking any changes in temperature. Mask wearing in the lab was implemented to protect individuals since workstation teams could not physically distance themselves while performing testing. BSL2 PPE was worn in the lab regardless of job responsibilities. Physical distancing was practiced outside of the lab when on breaks and eating meals; meals were provided onsite to the volunteers to further limit potential viral exposure.

Beyond the Clinical Lab. At full operational capacity, the UCSF Clinical lab extension involved approximately 120 personnel. As a result, access to an excess of human resources allowed us to recruit help for the local public health departments as they implemented large scale contact tracing and outbreak investigation activities.

M. Waste Management

Chemical and biohazard waste disposal services were established at the UCSF clinical lab extension as early as possible by coordinating with the Office of Environment, Health and Safety (EH&S) at UCSF. However, due to the rapid and urgent nature of bringing the clinical lab extension online, there was a gap between when operations commenced and when waste disposal services were established. To facilitate the effort, Chan Zuckerberg Biohub briefly acted as an intermediary and was initially responsible for disposing of chemical waste through their active UCSF EH&S account.

One of the key challenges that were addressed early on included standardization of waste disposal communication practices. As described in Section XX (Personnel, Volunteer Management), the volunteer squads were organized into discrete and dynamic units that could grow or shrink by the day in response to the estimated volume of work. As a result, it was not feasible to enact a centralized waste training process for all new recruits and it was impractical to assign waste disposal duties to a few members given the potential volume of chemical waste generated. It was also challenging to ensure that all volunteers were operating off of the latest information. The fluid nature of the environment resulted in occasional updates of waste disposal practices and sporadic communications from EH&S at UCSF. All of this information needed to be routed to each volunteer handling waste in a timely manner, so a tiered communication system was utilized.

To standardize communication, squad supervisors and several members from each squad were trained as waste disposal specialists and were walked through the chemical and biohazard waste disposal standard operating procedure. As specialists their responsibility was to ensure that their squad properly disposed of chemical and biohazard waste and they had the freedom to train other members as needed. However, new information and updates would only be propagated to the squad via these waste disposal specialists who would then distribute it to other members so that all volunteers were operating off of the most recent information. If squad members had questions or suggestions, they would interface with the specialists on their team and the specialists would interface with the waste management team as needed.

The chemical and biohazard waste disposal standard operating procedure was periodically revised to meet the needs of new sample testing protocols by synchronizing with squad supervisors and squad tech leads. To help steer the revisions, waste specialists also periodically collected feedback from their team regarding challenges and potential optimizations, which were largely incorporated. Prior to utilizing more efficient testing protocols and waste disposal practices, the UCSF clinical lab extension produced a peak output of more than 1000 gallons of solid waste per week.

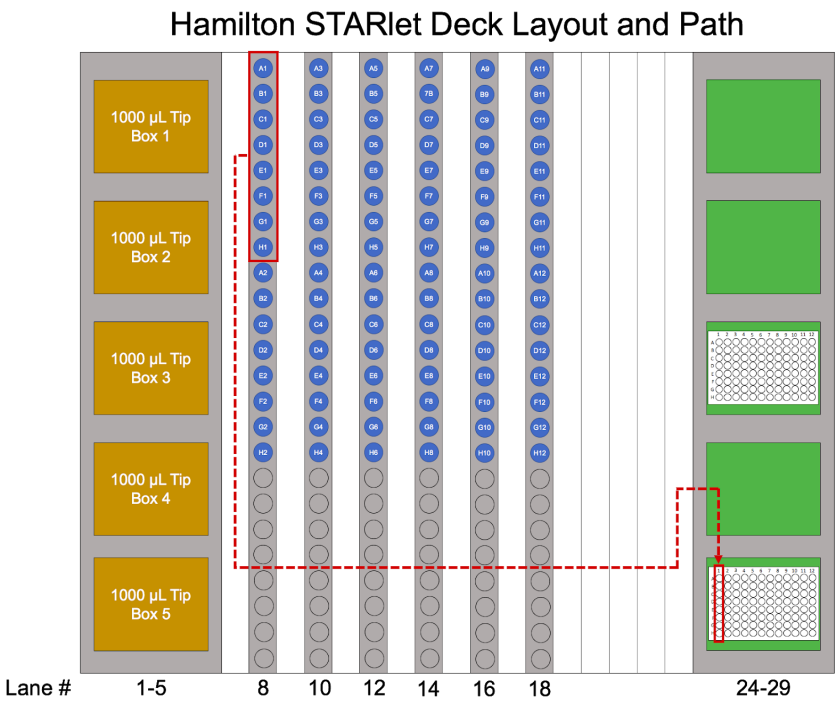
Summary/conclusions

1. Should acknowledge this effort was aided by funding we received but also discuss how a lot of the rules apply whether you are purchasing new equipment or not.

Supplementary Tables and Figures

Step	**# Cycles**	**Luna MM**	
		Temperature	**Time**
RT incubation	1	55 ºC	10 min
Enzyme activation	1	95 ºC	1 min
Amplification	45	95 ºC	10 sec
		58 ºC	30 sec

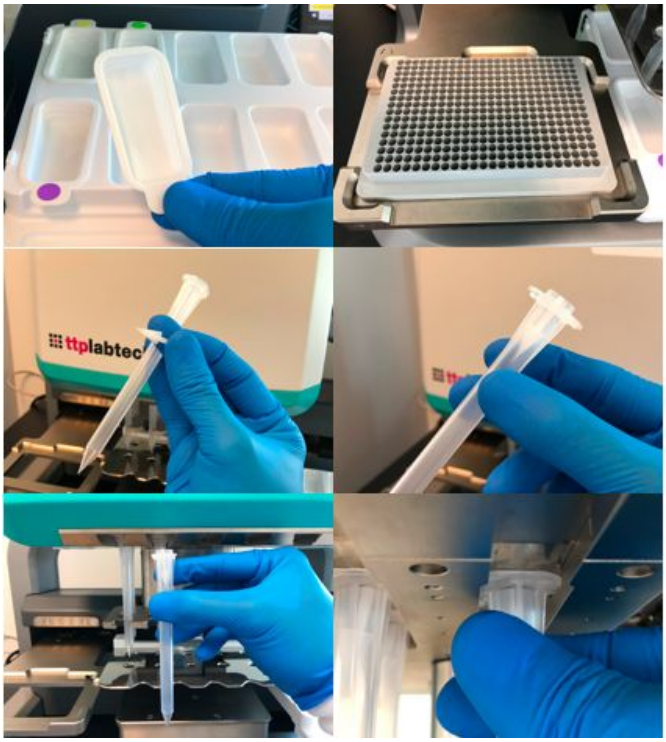
Supplementary Table X. RT-qPCR cycling conditions. MM: master mix.



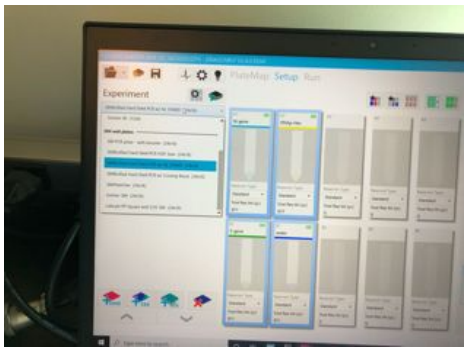
Supplemental FIGURE XX. Hamilton STARlet Deck Layout and Path. The 96 samples are loaded in 6 separate racks, with both empty lanes present between each rack and empty tube spaces at the end of each rack. These empty areas create a path (designated by the red dashed line) for pipette tips to avoid passing over any open tubes or non-target columns of the plate. Each tube rack contains two columns and the robot will transfer the samples sequentially through the columns.

Supplementary File X: DragonFly Set-Up & Operation

1. Obtain consumables and materials needed to operate the DragonFly: Aluminum Cold Block, DragonFly Syringes, Plungers and Reservoirs.
2. Outfit the DragonFly with selected consumables and materials (see picture below for guide). Ensure tip guard is down when inserting Syringes.

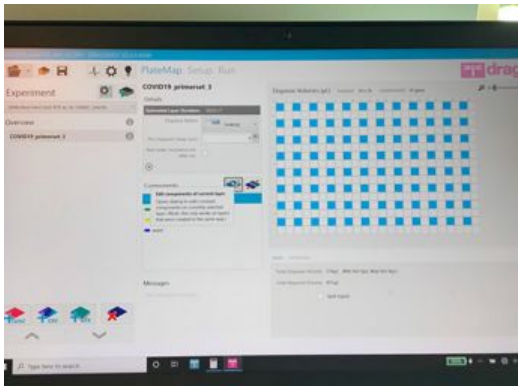


3. Open DragonFly software and the designated dispensing protocol.
4. Ensure 384well PCR plate is selected with the Aluminum Cold Block option as the plate definition.

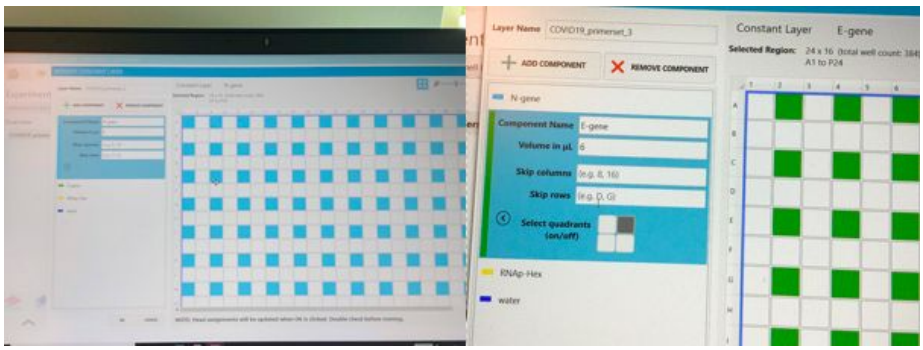


5. Check if the layer lists all of the components (Primer Mixes) to be used.
6. Under the “Plate Map” tab of the layer:

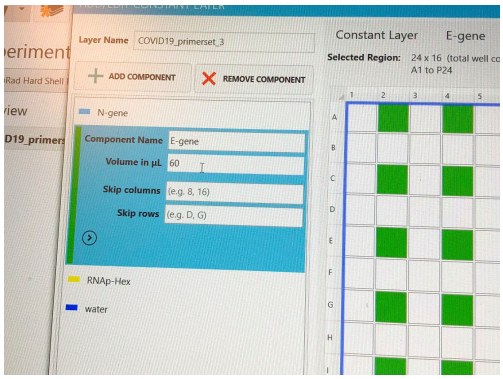
a. Click “edit components of the current layer”



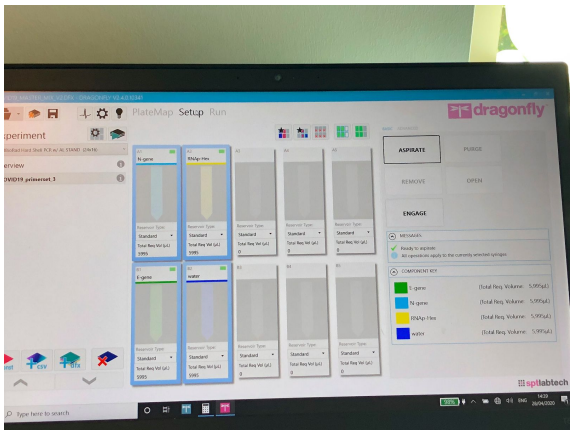
b. Edit component name, volume per well (6 uL) and quadrant position for each master mix used.



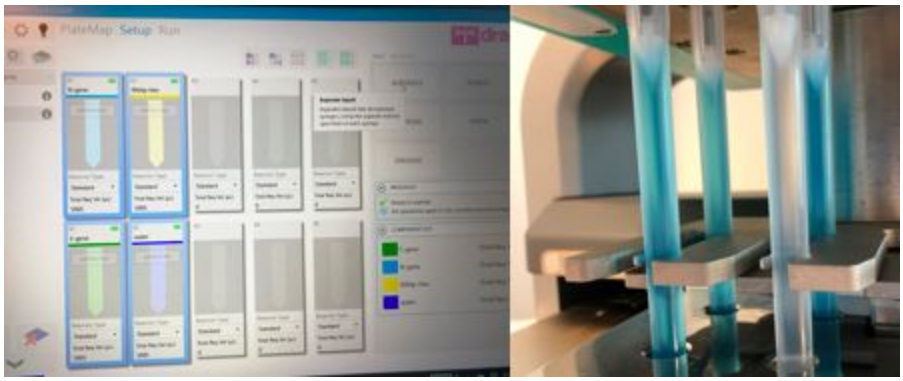
c. Ensure each syringe draws up the maximum volume (4000 uL) to avoid re-aspiration. One syringe volume’s worth of master mix dispenses about 7 plates (given 6uL/well).



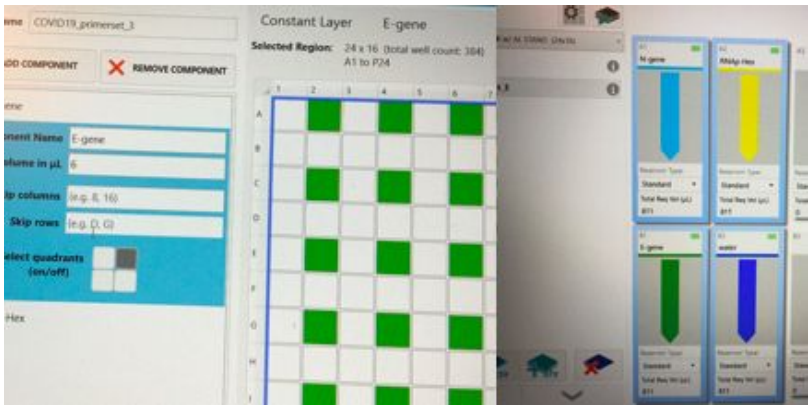
7. Once layer components (Primer Mixes) defined, click “SET UP” tab of the layer to match the syringe positions with the Primer Mixes.



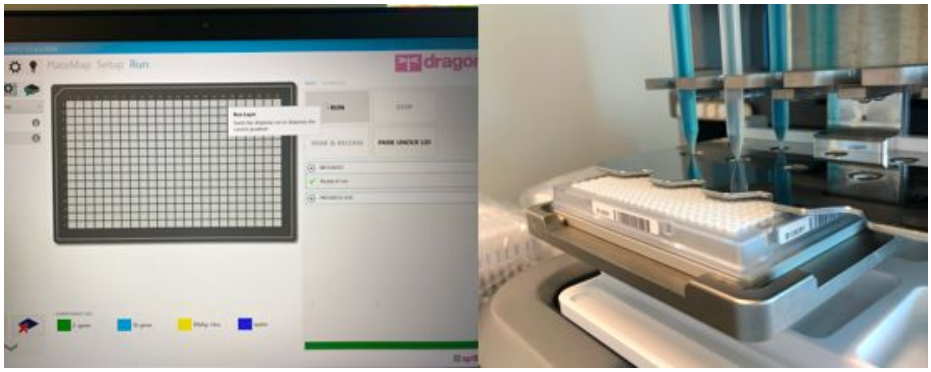
8. Click “Aspirate” function in the “SET UP” tab to aspirate the maximum volume of the syringe.



9. After aspiration, wipe the tip off the syringes to avoid discrepancy in volume dispensed.
10. CRITICAL Before running the protocol, go back to “Plate Map” tab and edit the component (Primer Mix) volume back to 6 uL.



11. Go to “RUN” tab of the layer and click “Run” to start the protocol.



12. Seal qPCR plate with Microseal F Seal to ensure protection from light.
13. Store qPCR plates in -20C.

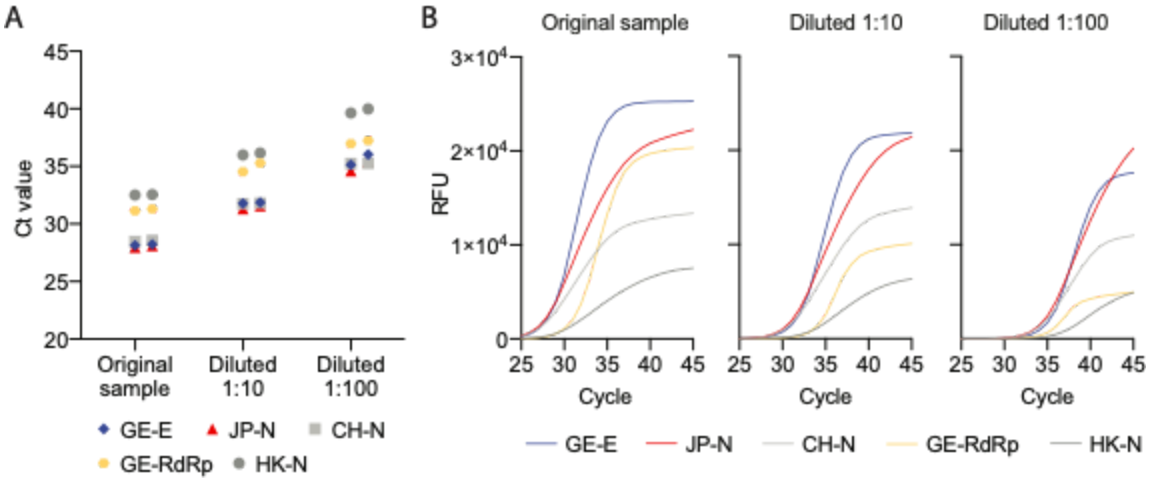


Figure SX. Comparison of (A) Ct values and (B) qPCR curves of SARS-CoV-2 RT-qPCR assays using viral culture extract. The data for two technical replicates are shown as individual points in panel A, and were combined into one average curve for panel B. Primer and probe concentrations were as described in Table X for GE-E, GE-RdRp and JP-N assays. The final primer concentration was 500 nM for the CH-N and HK-N assays. The final probe concentration was 250 nM for the HK-N probe and 200 nM for the CH-N probe. All assays were run using an extension temperature of 58ºC. GE: Germany, JP: Japan, CH: China, HK: Hong Kong, RFU: relative fluorescence units, Ct: threshold cycle.

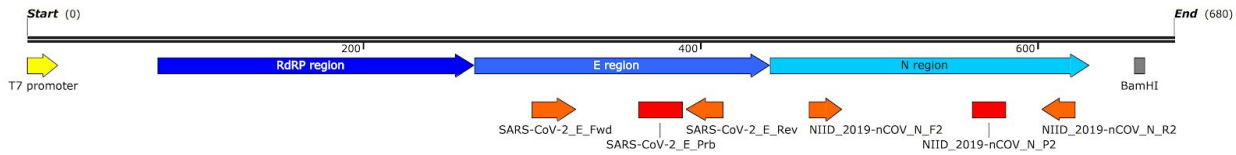


Figure SX. Plasmid map for *in vitro* transcribed RNA positive control template.

Supplementary data X. Sequence of insert for *in vitro* transcribed positive control RNA.
RdRp/E/N insert sequence:
GCTCGCAAACATACAACGTGTTGTAGCTTGTCACACCGTTTCTATAGATTAGCTAATGAGTGTGCTCA
AGTATTGAGTGAAATGGTCATGTGTGGCGGTTCACTATATGTTAAACCAGGTGGAACCTCATCAGGA
GATGCCACAACTGCTTATGCTAATAGTGTTTTTAACATTTGTCAAGCTGTCACGGCCAATGTTATGTA
CTCATTCGTTTCGGAAGAGACAGGTACGTTAATAGTTAATAGCGTACTTCTTTTTCTTGCTTTCGTGGT
ATTCTTGCTAGTTACACTAGCCATCCTTACTGCGCTTCGATTGTGTGCGTACTGCTGCAATATTGTTA
ACGTGAGTCTTGTAAAACCTTCTCGTGGTCCAGAACAAACCCAAGGAAATTTTGGGGACCAGGAACT
AATCAGACAAGGAACTGATTACAAACATTGGCCGCAAATTGCACAATTTGCCCCCAGCGCTTCAGCG
TTCTTCGGAATGTCGCGCATTGGCATGGAAGTCACACCTTCGGGAACGTGGTTGACCTACACAGGT
GCCATCAAATTGG